Lost in Twin-Field QKD Over 14 km to 1,000 km: Distance Records and Security Risks

Ranbir Singh [1], Aradhya Gupta [2], Shaivi Pathak [3], Rida Jangra[4]
Department of CSE-AIML, Bharati Vidyapeeth's College of Engineering, New Delhi

Abstract:

Twin-field quantum key distribution (TF-QKD) aims to make quantum communication more secure and efficient over long distances. This review looks at two 2026 studies: one demonstrating TF-QKD over a 14.2 km turbulent urban free-space link, and another examining security risks from manipulated reference beams. While the first shows promising key rates using phase estimation and channel post-selection, the second reveals potential attacks through intensity modulation and unmonitored wavelengths. Together, the studies highlight that improving communication range must go hand in hand with hardware security. However, these results do not yet demonstrate a complete satellite-based system. The review compares experimental findings, theoretical scaling, and security risks, without claiming new experimental results.

Introduction:

Quantum key distribution (QKD) allows two people to share secret keys and detect possible eavesdropping. However, sending quantum signals over long distances is difficult because the signals weaken as they travel. Twin-field QKD (TF-QKD) helps overcome this problem and improves communication over longer distances.

Two recent studies explore its progress and challenges. One demonstrates TF-QKD over a 14.2 km urban free-space link, while the other investigates attacks that expose weaknesses in the system's hardware. This review examines both studies to understand how TF-QKD can support long-distance communication while remaining secure.

Acknowledgement

We sincerely thank Dr. Yugnanda Malhotra, Dean (Research and Development), and Dr. Prateek Varshney, Professor, Department of Physics, for their guidance and support throughout this work. We are grateful to Dr. Dharmendra Saini, Principal, Bharati Vidyapeeth's College of Engineering, New Delhi, for providing the platform to present it. Finally, we thank the scientific community, whose research made this review possible.

Theory:

What are the basics?
Alice and Bob are two standard names for the users, Charlie is the middle measurement station. Weak coherent pulses are heavily attenuated laser pulses. Decoy states are pulses sent at different intensities so the users can bound photon-number effects without assuming perfect single-photon sources.

What is the protocol flow?
Alice and Bob each send phase-encoded weak pulses to Charlie. Charlie interferes with them on a beam splitter, measures with single-photon detectors, and announces the outcome. Alice and Bob then sift the events that count toward the key, then estimate errors using decoy-state analysis, reconcile raw keys (error correction) and, apply privacy amplification to reduce any adversary's information
A detector click is not yet a secret key; security comes from the full chain.

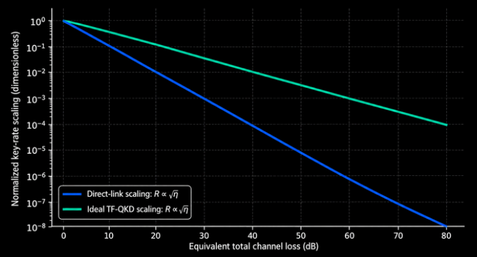
Why can Charlie be untrusted?
TF-QKD is measurement-device-independent (MDI), the security proof binds what an adversary learns from Charlie's announcements and the observed statistics. Source flaws, side channels, and software errors are not automatically removed, so the protocol's assumptions must match the real hardware.

Why Loss Scaling Matters?
With channel transmittance η (0 = no transmission, 1 = no loss),
loss in dB is L = −10 $\log_{10}(\eta)$. For example, 20 dB loss means η = 0.01.
Direct QKD: rate ∝ η TF-QKD (ideal): rate ∝ √η
This is a scaling comparison, not a numerical promise: real rates depend on noise, finite-key effects, and protocol penalties. As loss grows, √η and η separate rapidly

Suggested validation workflow for future TF-QKD research



The scaling advantage of twin-field QKD under increasing loss



Secure Key Rate vs. Transmission Distance in TF-QKD

Distance	Key rate
499 km	47.9 bit/s
701 km	0.98 bit/s
800 km	0.051 bit/s
952 km (finite-size, positive key)	0.0031 bit/s
1002 km (asymptotic only)	0.0034 bit/s

Future Research

Build a reproducible testing framework that measures channel stability, key generation and optical monitoring under the same conditions.
Test defenses against reference-beam attacks over realistic free-space links. Study atmospheric channel variation and reference-beam attacks together, in one experimental setup.
Extend TF-QKD toward satellite-based communication.Run a systematic literature search to confirm the research gap is novel.

Limitations:

- Recent studies have improved long-distance TF-QKD and exposed practical security risks, but the two are usually examined separately.
- How atmospheric channel variation and reference-beam attacks interact has not been studied together.
- To our knowledge, no study has addressed this combination.
- Focused literature review, not a systematic one; no database search was done.
- The experimental graph uses hourly samples from data recorded at shorter intervals, so it does not reproduce the full datasets.
- Theoretical plots use simplified laws that exclude noise and detection efficiency, so they are illustrative, not predictive.
- The distance comparison mixes different media and conditions, so it is not a direct ranking.

Discussion:

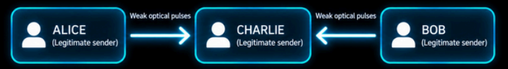
The study on TF-QKD proposes a way of achieving longer distance quantum communication while maintaining a high level of security. Both studies tackle different aspects of the task, as the first one attempts to find a practical implementation, while the second one tries to address the security aspect.

The free-space experiment featured a 14.2 km urban atmosphere channel, over which the TF-QKD protocol was tested. The authors utilized stabilized laser sources, which featured timing and phase estimation techniques to counteract the challenges posed by the atmospheric turbulence. A channel-efficiency post-selection technique was used to improve the ratio of the two channels, as the imbalances in them decrease the key generation rate.

The post-selection process has proven beneficial for the key generation in the case of TF-QKD, as the value increased from 2.178 Mbits at 44.1 bits/s to 4.980 Mbits at 100.8 bits/s. However, the overall usage of the channel was decreased due to a lower ratio of useful data. The results of the first study have demonstrated that a proper choice of data samples can increase the amount of data retrieved from a quantum channel, while the channel conditions should also be taken into account.

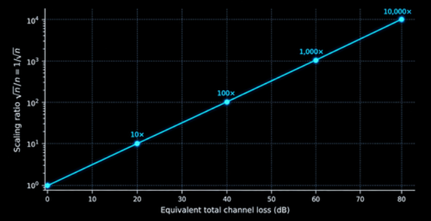
Meanwhile, the second study addressed a different aspect of TF-QKD, namely, the possibility of attacks. The authors have demonstrated that the ability to modify reference beams with intensity fluctuations on a short timescale or at wavelengths beyond the detection range can create new security vulnerabilities.

In summary, the two studies address different aspects of TF-QKD's utilization. The first study, which is a practical experiment, demonstrates that stable and long-haul communication can be achieved with proper hardware selection and additional data processing. On the other hand, the second study shows that the security of such a system cannot be taken for granted and may have loopholes due to various sources of errors. Therefore, the future research should focus on finding practical ways of increasing transmission distance and ensuring that the security level of TF-QKD is sufficient for its practical implementation, possibly in the case of free-space and satellite communications.



Single-photon interference + detector announcement

Why the scaling difference matters at high loss



Conclusion

Twin-field quantum key distribution (TF-QKD) has the potential to make secure communication over long distances more efficiently. One study demonstrated TF-QKD over an urban atmospheric link, achieving promising secure key rates. The results also showed how atmospheric changes, signal strength and phase stability can affect communication.

Another study demonstrated attacks that exploit weaknesses in reference-beam monitoring and may compromise system security. Together, these studies show that improving communication distance is only part of the challenge. Reliable hardware, proper monitoring and protection against practical attacks are equally important. Future research should focus on testing these challenges under realistic conditions to make quantum communication more secure and reliable.



References:

[1] Y.-H. Li et al., "Free-space twin-field quantum key distribution," Jul. 2026.
[2] S. Juárez et al., "Reference-beam attacks against twin-field quantum key distribution using optical injection locking," Mar. 2026.
[3] S. M. Arslan et al., "Twin-field quantum key distribution: Protocols, security, and open problems," Oct. 2025.
[4] S. Wang et al., "Twin-field quantum key distribution over 830-km fibre," Feb. 2022.
[5] Y. Liu et al., "Experimental twin-field quantum key distribution over 1000-km fiber distance," May 2023.